Understanding Artificial Intelligence

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What is AI, and how does it compare to normal programs?

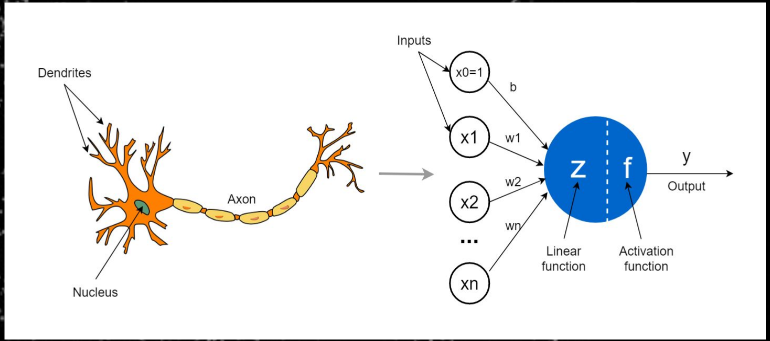
- Attempt to replicate human cognition
- Logic exists mostly as a state of data
- Ever changing "black box"
- Driven by patterns in data

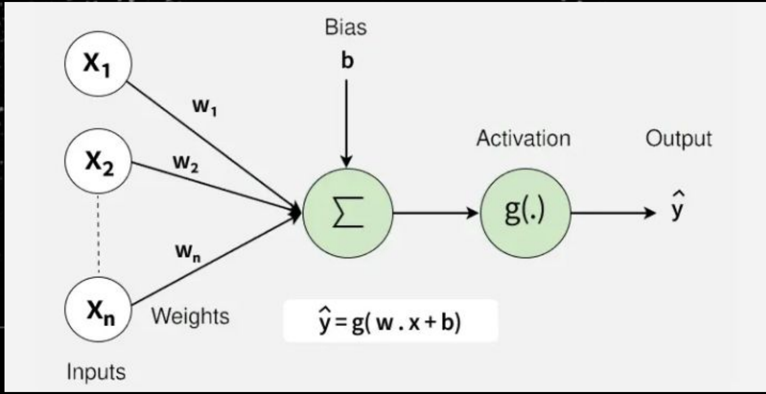
How does AI relate to how our own brains function?

- Brain is mostly box of neurons
- Neurons as chains of varying impulses
- We take input from our world and learn from it
- Neural Networks use neurons to process inputs

Neurons and Artificial Neurons

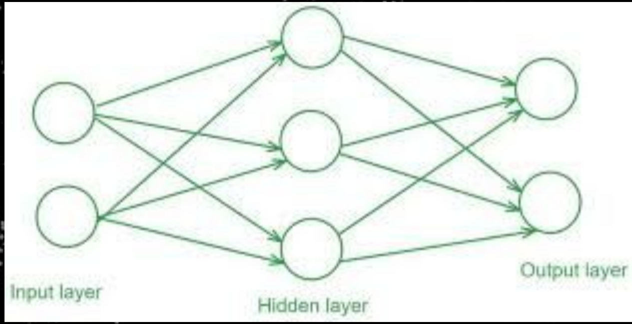
- Simplification of a neuron
- Neuron takes multiple inputs
- Produces constant output
- Not exactly same
- Math behind one neuron





Neural Networks

- Consists of layers of artificial neurons
- Mimics how we summarize detailed data
- Each layer is dependant on previous one
- Gradient descent



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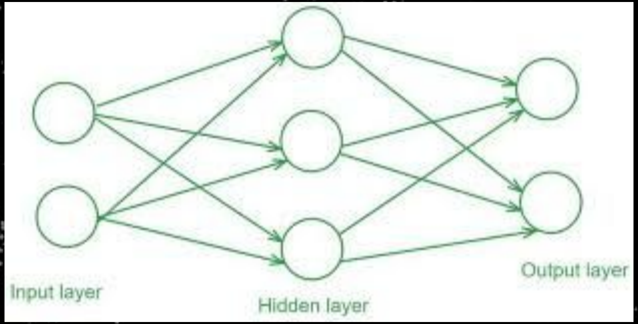
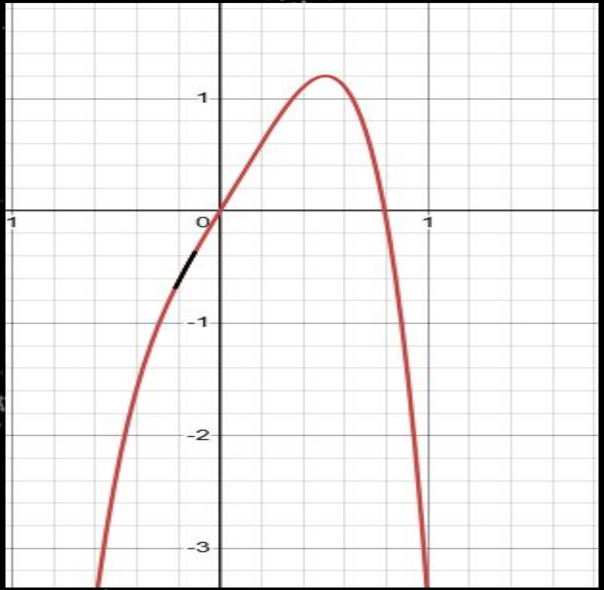


Back propagation & Gradient Descent

- Start off with an output
- Get function of Loss
- Calculate gradient for each layer
- Optimize weights based



Neural Networks are not enough

- Cannot understand context by itself
- How should we deal with varying input/output sizes?

Transformers

- Basic networks lack context and order
- Transformers process the whole input at once
- Foundation of GPT, Claude, Gemini

Attention mechanisms

- How relevant is each part of the input to every other part?
- Every token scores against every other token
- High score = pay more attention

Sources

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